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#include <stdio.h>

#include <stdlib.h>

// Node structure
struct Node {
    int data;
    struct Node* left;
    struct Node* right;
};

// Function to create a new node
struct Node* createNode(int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->left = newNode->right = NULL;
    return newNode;
}

// Function to insert into BST
struct Node* insert(struct Node* root, int data) {
    if (root == NULL) {
        return createNode(data);
    }
    if (data < root->data) {
        root->left = insert(root->left, data);
    } else if (data > root->data) {
        root->right = insert(root->right, data);
    }
    return root;
}

// In-order Traversal (Left, Root, Right)
void inorder(struct Node* root) {

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    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}

// Pre-order Traversal (Root, Left, Right)
void preorder(struct Node* root) {
    if (root != NULL) {
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}

// Post-order Traversal (Left, Right, Root)
void postorder(struct Node* root) {
    if (root != NULL) {
        postorder(root->left);
        postorder(root->right);
        printf("%d ", root->data);
    }
}

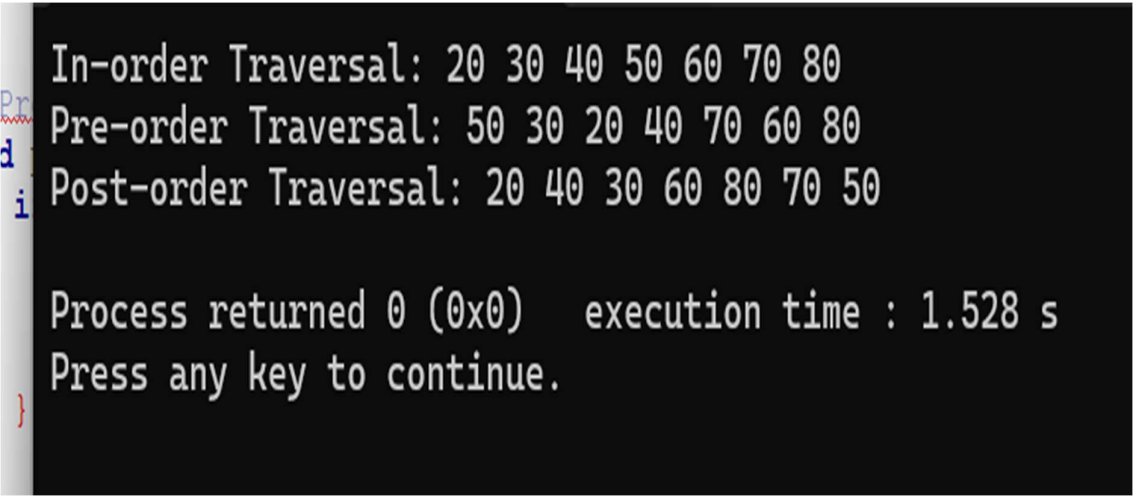
// Main function
int main() {
    struct Node* root = NULL;

    // Construct BST
    root = insert(root, 50);
    root = insert(root, 30);
    root = insert(root, 70);

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    root = insert(root, 20);  
    root = insert(root, 40);  
    root = insert(root, 60);  
    root = insert(root, 80);  
    printf("In-order Traversal: ");  
    inorder(root);  
    printf("\n");  
    printf("Pre-order Traversal: ");  
    preorder(root);  
    printf("\n");  
    printf("Post-order Traversal: ");  
    postorder(root);  
    printf("\n");  
    return 0;  
}
```

OUTPUT:



```
In-order Traversal: 20 30 40 50 60 70 80  
Pre-order Traversal: 50 30 20 40 70 60 80  
Post-order Traversal: 20 40 30 60 80 70 50  
  
Process returned 0 (0x0)   execution time : 1.528 s  
Press any key to continue.
```